

Program 5: Demonstrate CI integration by running jenkins in a docker container

File structure:

Program-5

/Dockerfile

/docker-compose.yml

1) Create folder: >> mkdir lab5

>> cd lab5

2) Dockerfile

```
lrv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program5$ cat Dockerfile
FROM jenkins/jenkins:lts

# Switch to root user to install tools
USER root

# Install docker inside jenkins container
RUN apt-get update && apt-get install -y docker.io

# Switch back to Jenkins user
USER jenkins
```

3) docker-compose.yml

```
lrv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program5$ cat docker-compose.yml
version: '3.8'

services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

4) Build and run the containers

Using Docker File

#sudo docker build -t lab5 .

#sudo docker run -p 8080:8080 -p 50000:50000 --name jenkins_ci
lab5

```

1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms$ cd Program5
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program5$ touch Dockerfile
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program5$ open Dockerfile
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program5$ touch docker-compose.yml
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program5$ open docker-compose.yml
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program5$ docker compose build
WARN[0000] /home/1rv24mc031_darshan_hegde/Desktop/LabPrograms/Program5/docker-compose.yml: the attribute `version` is o
confusion
[+] Building 10.6s (8/8) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 577B                                          0.0s
=> [internal] load build definition from Dockerfile                    0.0s
=> => transferring dockerfile: 250B                                    0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:latest    10.5s
=> [internal] load .dockerignore                                       0.0s
=> => transferring context: 2B                                          0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:latest@sha256:7b1c378278279c8688efd 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io    0.0s
=> exporting to image                                                 0.0s
=> => exporting layers                                                0.0s
=> => writing image sha256:671b2f986cdce9b58c518ffeb2aee460caa1814127929 0.0s
=> => naming to docker.io/library/program5-jenkins                   0.0s
=> resolving provenance for metadata file                             0.0s
[+] Building 1/1
✔ program5-jenkins Built                                             0.0s

```

5)#sudo docker compose up

```

1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program5$ docker compose up
WARN[0000] /home/1rv24mc031_darshan_hegde/Desktop/LabPrograms/Program5/docker-compose.yml: the attribute `version` is obsole
confusion
[+] Running 1/1
✔ Container jenkins-ci Created                                         0.0s
Attaching to jenkins-ci
jenkins-ci | Running from: /usr/share/jenkins/jenkins.war
jenkins-ci | webroot: /var/jenkins_home/war
jenkins-ci | 2025-11-21 04:20:10.012+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war f
jenkins-ci | 2025-11-21 04:20:10.075+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contex
jenkins-ci | 2025-11-21 04:20:10.125+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; buil
ae1d1d333ce2ea; jvm 21.0.9+10-LTS
jenkins-ci | 2025-11-21 04:20:10.525+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP S
pServlet
jenkins-ci | 2025-11-21 04:20:10.580+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=
jenkins-ci | 2025-11-21 04:20:10.923+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory
KINS_HOME")
jenkins-ci | 2025-11-21 04:20:11.030+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.Conte
ile:///var/jenkins_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@6d467c87{STARTED}}
jenkins-ci | 2025-11-21 04:20:11.040+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnec
jenkins-ci | 2025-11-21 04:20:11.051+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server
jenkins-ci | 2025-11-21 04:20:11.052+0000 [id=34] INFO winstone.Logger#logInternal: Winstone Servlet Engine running
jenkins-ci | 2025-11-21 04:20:11.149+0000 [id=33] INFO jenkins.model.Jenkins#<init>: Starting version 2.528.2
jenkins-ci | 2025-11-21 04:20:11.214+0000 [id=41] INFO jenkins.InitReactorRunner$1#onAttained: Started initializati
jenkins-ci | 2025-11-21 04:20:11.262+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
jenkins-ci | 2025-11-21 04:20:11.903+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
jenkins-ci | 2025-11-21 04:20:11.912+0000 [id=44] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
jenkins-ci | 2025-11-21 04:20:11.913+0000 [id=43] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extens
jenkins-ci | 2025-11-21 04:20:12.145+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
jenkins-ci | 2025-11-21 04:20:12.145+0000 [id=42] INFO jenkins.InitReactorRunner$1#onAttained: System config adapte

```

6) Go to browser and type <http://localhost:8080> to see the output
See the list of containers and volumes which were created

